

STARBIRTH

STARS FORM OUT OF VAST CLOUDS OF COOL GAS AND DUST, CALLED MOLECULAR CLOUDS, THAT OCCUPY PARTS OF INTERSTELLAR SPACE. THE PROCESS OF STAR FORMATION WITHIN THESE CLOUDS CAN TAKE MILLIONS OF YEARS.

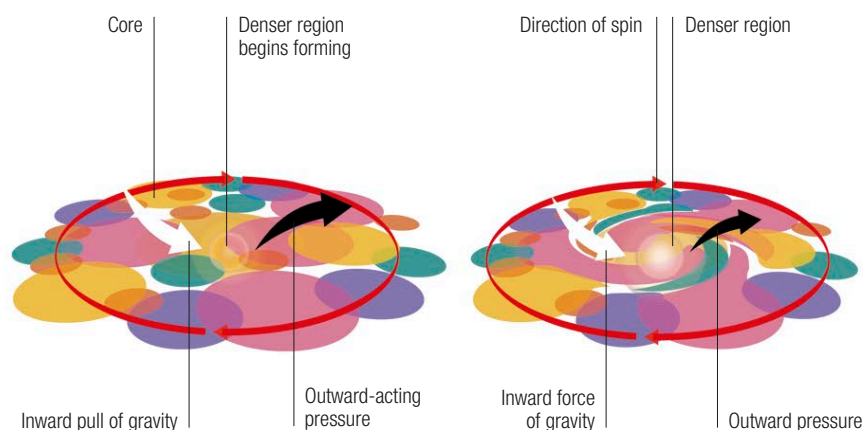
The molecular clouds where stars are born can be hundreds of light-years across. Most sites of star formation are hidden inside these dense dusty clouds. However, there are places where the radiation from brilliant newly formed stars is clearing the dust away and is lighting up the surrounding gas. We see these star-forming regions as bright nebulae. They include the Eagle Nebula (see opposite) in Serpens, the Orion Nebula (see pp.164–65), and many others. Some specific dark concentrations of dust and gas sometimes seen within molecular clouds are known as Bok globules. These frequently result in the formation of double or multiple star systems (see pp.40–41).

Star formation

For star formation to start within a molecular cloud, a triggering event is needed. This could be a nearby supernova explosion, the passage of the cloud through a more crowded region of space, or an encounter with a passing star. The tidal forces and pressure waves that come into action during these situations push and pull at the cloud, compressing parts until some regions become dense enough for stars to form. Gravity then does the rest of the work of forming each star, pulling more and more material onto the developing knot of matter and concentrating most of it at the center. As the material grows denser, random motions are transformed into a uniform spin around a single axis. Collisions between particles jostling within the cloud raise its temperature, notably in the center, and the newly forming star begins to glow with infrared (heat) radiation.

At this stage, the protostar (newly forming star) is quite unstable. It loses mass by expelling gas and dust, directed in two opposing jets from its poles. At its center, it eventually becomes so hot that nuclear fusion starts, and as the balance between gravity and outward-acting pressure begins to equalize, the protostar settles down to become a main-sequence star.

Astronomers have calculated that on average about **seven new stars per year** are born in the Milky Way Galaxy, most of them somewhat **smaller than the Sun**

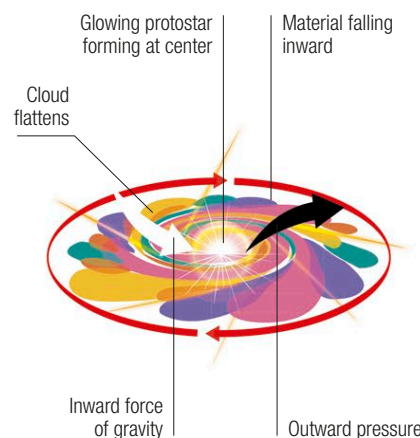


1 Dense region forms in a molecular cloud

Some nearby event, such as a supernova, causes dense clumps to come together inside a molecular cloud under the action of gravity. These clumps will become clusters of stars. They break up further into smaller regions called cores.

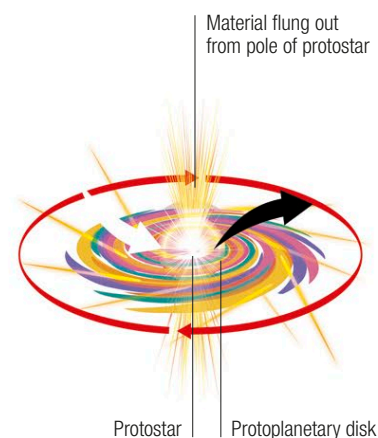
2 Core starts to collapse

Each core then starts to contract under the influence of gravity, and begins to slowly spin. Over tens of thousands of years, this spinning, gradually concentrating mass of gas and dust collapses down to less than a light-year across.



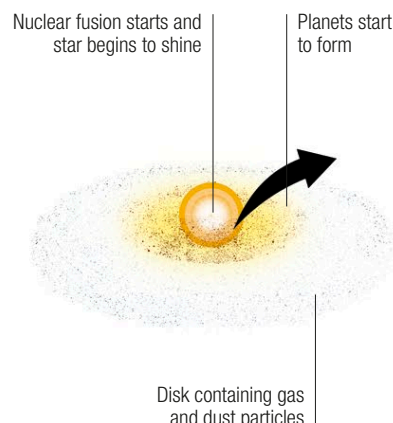
3 Protostar forms

The contracting cloud forms into a flattened, spinning disk, a few light days across, with a hot central bulge, which eventually stabilizes as a rapidly spinning protostar. Material from the cloud falls inward and feeds onto the star.



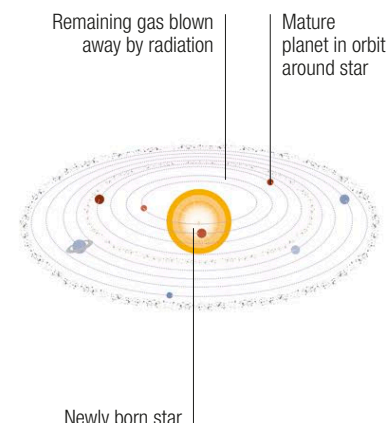
4 Protostar ejects material from its poles

Eventually the protostar spins so rapidly that new material falling onto it is flung back off. This excess material forms two tight jets emerging along the rotation axis. The cloud around the protostar flattens to form a protoplanetary disk.



5 Star ignites

When its central core becomes hot enough, nuclear fusion reactions start within the protostar and it begins to shine as a fully fledged star. Over millions of years, planets may gradually grow from material in the disk of dust and gas.



6 Planetary system forms

Radiation pressure from the newborn star blows away the remaining gas (some may accrete onto gas giant planets). Eventually, all that remains is the star, any planets, and possibly some smaller bodies, such as comets and asteroids.